

AJAY CARPENTER

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SUMMARY

B.Tech student in Artificial Intelligence and Data Science with strong foundations in Python, OOP, Machine Learning, and Data Science. Hands-on experience developing AI/ML projects involving NLP, Computer Vision, and model training and evaluation. Familiar with NumPy, Pandas, Scikit-learn, TensorFlow, and Flask. Analytical and Curious about emerging technologies with a strong willingness to learn, grow, and apply new concepts to real-world AI/ML problems.

TECHNICAL SKILLS

- **Programming and APIs:** Python, SQL, OOP, Fast API
- **Machine Learning & AI:** Machine Learning, Deep Learning, Generative AI, NLP, Model Training & Evaluation, LLMs, Prompt Engineering
- **Data Science:** NumPy, Pandas, Scikit-learn, Data Preprocessing, Data Analysis
- **Computer Vision:** TensorFlow.js, OpenCV, Image Processing
- **Tools:** Flask, Git, GitHub, Flutter, Docker, MS Fabric

PROJECTS

SmartCart AI | *React, Flask, Scikit-Learn, NLP*

- Engineered a React and Flask e-commerce platform featuring ML-based price prediction and fake review detection, leveraging AI-assisted prompt engineering to accelerate development velocity and improve user shopping efficiency by 50%.
- Set up live cloud infrastructure by securely linking a MongoDB Atlas database via environment variables, syncing GitHub code to host the backend on **Render**, and wiring the **API** into a live front-end deployment on **Vercel**.

SafeGuard AI | *React, TensorFlow.js, MongoDB*

- Built a real-time face mask detection platform using Node.js and TensorFlow.js, utilizing iterative AI-assisted generation workflows to achieve a 94% model accuracy rate while reducing edge-based browser inference latency.
- Configured the end-to-end cloud pipeline from GitHub repository to **Render** container hosting, injecting MongoDB Atlas connection strings and security keys to serve a live application globally via **Vercel**.

QuizIt | *React, Supabase, Fast API*

- Architected a real-time gamified testing platform across 7 developer domains, using prompt engineering pipelines to dynamically generate and validate technical questions into structured JSON.
- Engineered an adaptive difficulty engine, backed by Supabase authentication and a real-time database to power secure profiles and live global leaderboards.

CERTIFICATIONS

- Data Analytics Job Simulation — Deloitte (July 2026): Completed practical industry tasks covering exploratory data analysis, data quality assessment, and forensic technology workflows via Forage.

EDUCATION

Sagar Institute of Research and Technology-Excellence, Bhopal

Aug 2023 – May 2027

B.Tech – Artificial Intelligence and Data Science | SGPA 7.5 (5th sem)